

- AI is building systems that simulate intelligent behavior.
- ML is a subset of AI that makes decisions by fitting mathematical models to seen data.

## 1.1: Supervised Learning:

- Defines a mapping between input data and output predictions.
- Regression Problems produce continuous output
  - ↳ Prices, ages, i.e. no pre defined classes
  - ↳ Multivariate regression outputs more than 1 number
- Classification Problems assign the input to classes
  - ↳ object detection, recognition, etc.
  - ↳ Binary [ $N = 2$  categories]
  - ↳ Multiclass [ $N > 2$  categories]
- Inputs matter!

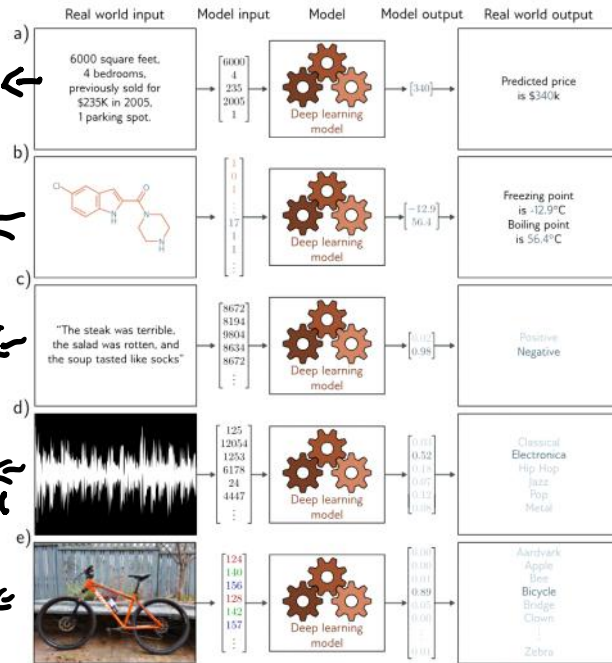
Tabular, constant length  
no internal structure

Geometry of molecule  
+  
Atoms in it matter

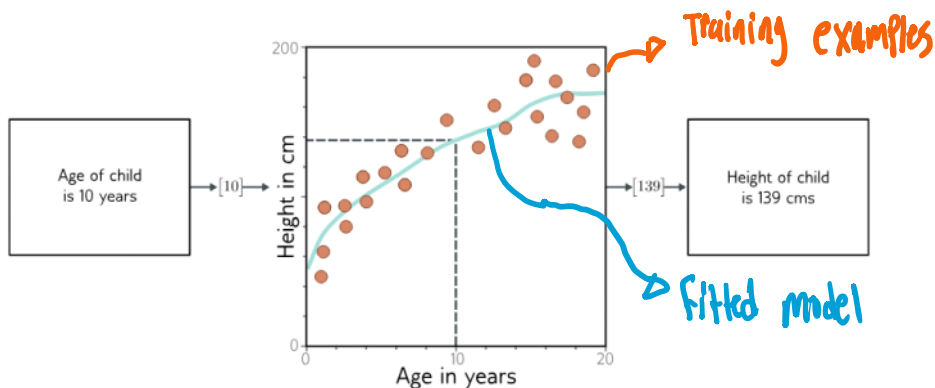
Variable length,  
order matters!

Very high dimensional  
model needs to process sizable  
inputs

Has spatial structure



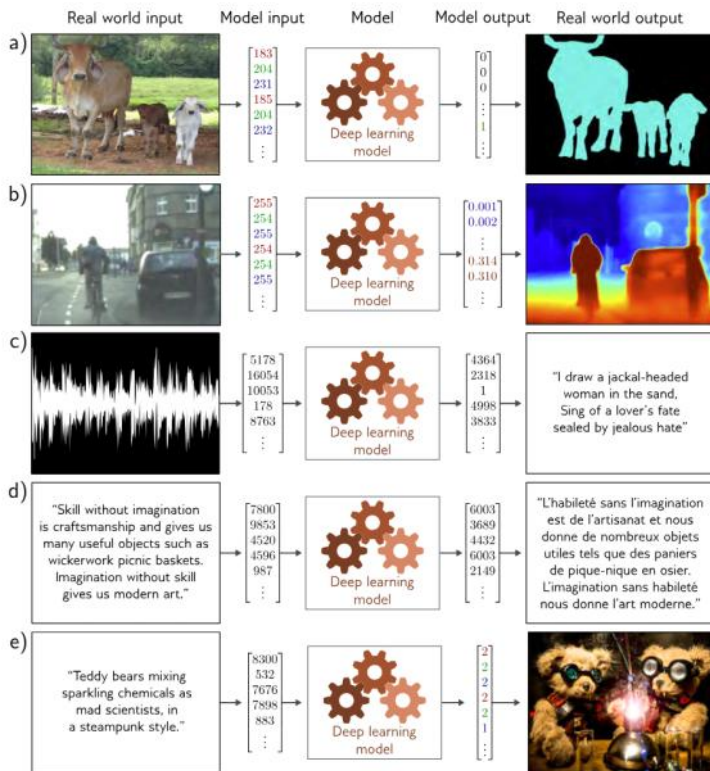
- A model represents a family of equations mapping input to output.
- The resultant equation is found using the training data
- Training the model means searching through this family of equations and finding the equation that best describes the training data.



- Deep Neural Networks can handle the above nuances
  - ↳ Very large inputs
  - ↳ Variable length inputs
  - ↳ Various internal structures

↳ Various internal structures

- Additionally, the outputs can be structured.



Output is high dimensional & structured. The structure is closely tied to the input.  
 (If an output pixel belongs to cow Odds are the neighboring pixel belongs to cow)

Output has a complex structure that isn't tied to the input.  
 The output can be ambiguous: Multiple valid English → French translations

The output is very structured: not all strings of words make valid English sentences.

- We have to learn the mapping + "grammar" structure of outputs.

- Deep neural networks can learn the structure without dedicated labels.

## 1.2: Unsupervised Learning:

- Constructing a model from input data without corresponding output labels

↳ The goal is understanding / describing the structure of the data.

↳ NOT producing a mapping!!!

- Data can be in various forms (discrete, continuous / low or high dimensional / variable length)

- The book focuses on nonrecursive / nonparametric methods

- The book focuses on generative unsupervised models.  
 ↳ Learn to generate data that is statistically indistinguishable from the training data

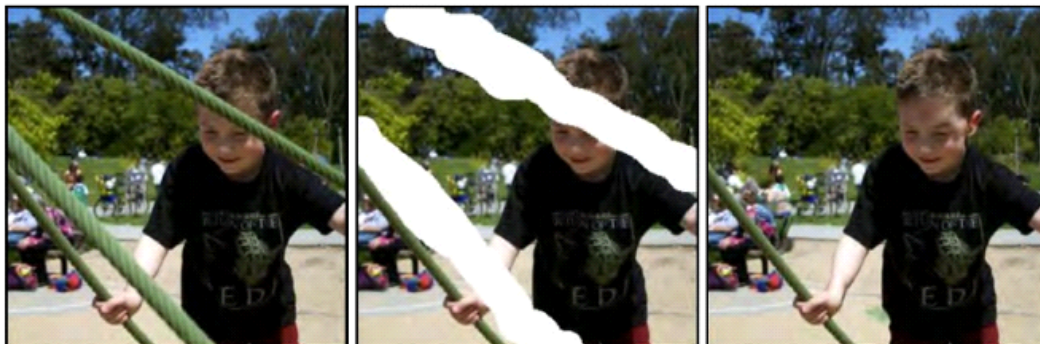


**Figure 1.5** Generative models for images. Left: two images were generated from a model trained on pictures of cats. These are not real cats, but samples from a probability model. Right: two images generated from a model trained on images of buildings. Adapted from Karras et al. (2020b).

The moon had risen by the time I reached the edge of the forest, and the light that filtered through the trees was silver and cold. I shivered, though I was not cold, and quickened my pace. I had never been so far from the village before, and I was not sure what to expect. I had been walking for hours, and I was tired and hungry. I had left in such a hurry that I had not thought to pack any food, and I had not thought to bring a weapon. I was unarmed and alone in a strange place, and I did not know what I was doing.

I had been walking for so long that I had lost all sense of time, and I had no idea how far I had come. I only knew that I had to keep going. I had to find her. I was getting close. I could feel it. She was nearby, and she was in trouble. I had to find her and help her, before it was too late.

**Figure 1.6** Short story synthesized from a generative model of text data. The model describes a probability distribution that assigns a probability to every output string. Sampling from the model creates strings that follow the statistics of the training data (here, short stories) but have never been seen before.



**Figure 1.7** Inpainting. In the original image (left), the boy is obscured by metal cables. These undesirable regions (center) are removed and the generative model synthesizes a new image (right) under the constraint that the remaining pixels must stay the same. Adapted from Saharia et al. (2022a).

- Some generative models exploit the fact that the training data can be

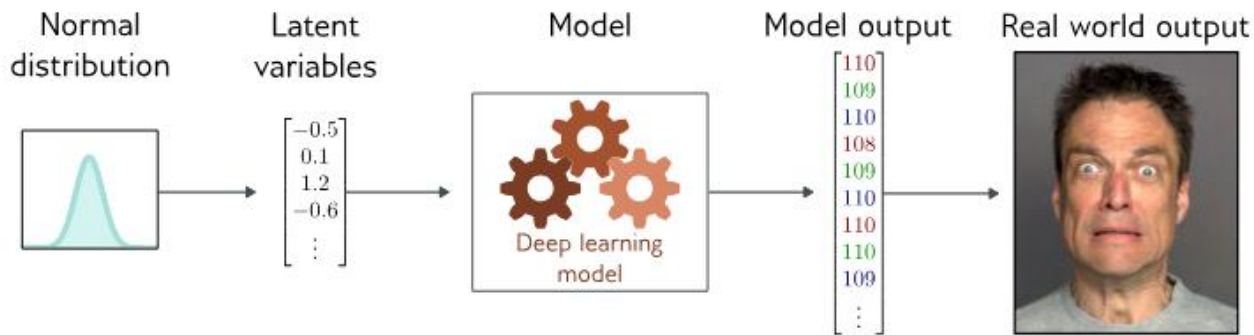


- Some generative models exploit the fact that the training data can be represented with lower dimensional features: these are called latent variables.

↳ Ex: A  $32 \times 32$  image of a cylinder is described in  $3 \times 32 \times 32$  dimensions.

In reality to produce an image we need 2 latent variables: height; width.

The role of deep learning is to describe the mapping between the data ; the latent variables.



**Figure 1.10** Latent variables. Many generative models use a deep learning model to describe the relationship between a low-dimensional “latent” variable and the observed high-dimensional data. The latent variables have a simple probability distribution by design. Hence, new examples can be generated by sampling from the simple distribution over the latent variables and then using the deep learning model to map the sample to the observed data space.



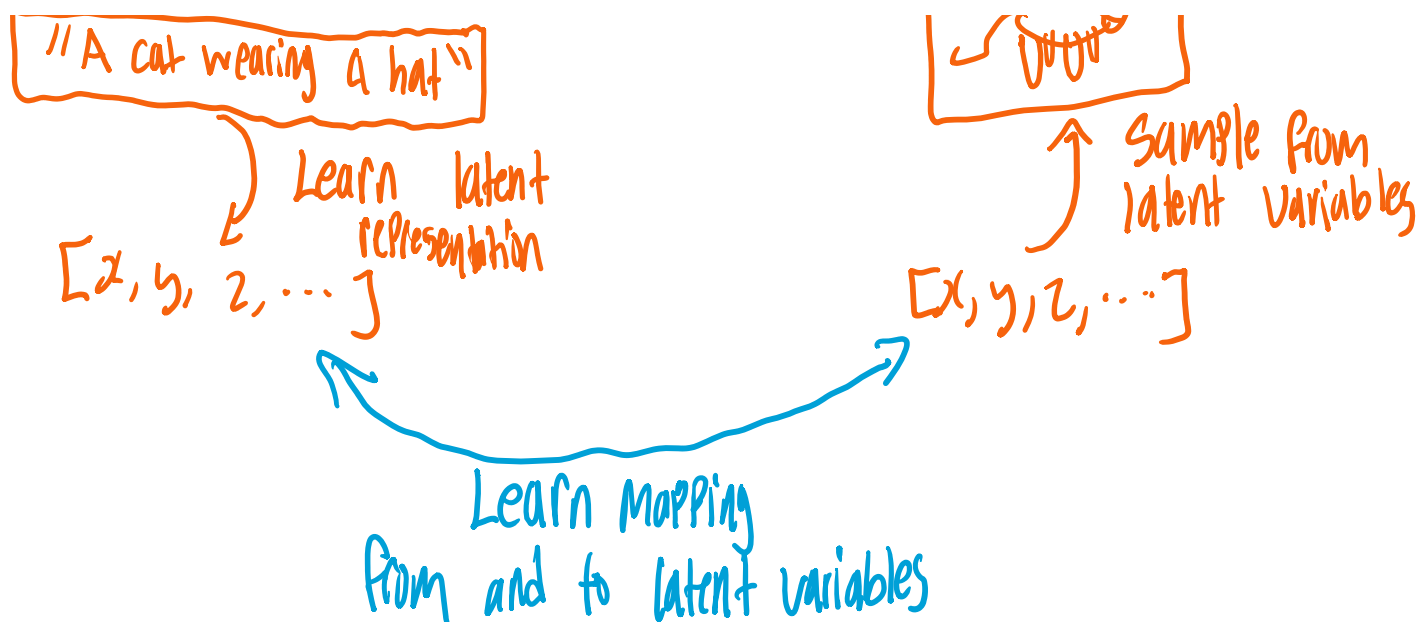
**Figure 1.11** Image interpolation. In each row the left and right images are real and the three images in between represent a sequence of interpolations created by a generative model. The generative models that underpin these interpolations have learned that all images can be created by a set of underlying latent variables. By finding these variables for the two real images, interpolating their values, and then using these intermediate variables to create new images, we can generate intermediate results that are both visually plausible and mix the characteristics of the two original images. Top row adapted from Sauer et al. (2022). Bottom row adapted from Ramesh et al. (2022).

• We can connect unsupervised & supervised models. For example:

↳ Caption to image generation:

"A cat wearing a hat"





~~Un~~ = Unsupervised  
~~Sup~~ = Supervised

• This has 3 advantages:

- 1) Need fewer text/image pairs since its lower dimensional
- 2) More likely to generate correct images, just needs sensible latent variable values
- 3) we can add randomness to the mapping to produce multiple correct images.

### 1.3: Reinforcement Learning:

- An agent which lives in an environment & learns by interacting with it.
- The goal is for the agent to choose actions that maximize rewards, on average.
- Not simple; trade off between exploration & exploitation

- One application of deep learning is to approximate the optimal policy for the agent.

## 1.4: Ethics:

- Five concerns (not exhaustive):

### 1) Bias ; Fairness:

Training a model to predict salary based on historical data will result in the system inheriting historical bias.

↳ Will predict women should be paid less than men.

### 2) Explainability:

Systems are very large; we don't know why the system makes the decisions it does.

↳ some success in producing local explanations → Look up, this sounds very interesting. Grennan, et al.

### 3) Weaponization:

This is already happening.

↳ Heikkilä, 2022

### 4) Concentrating power:

Companies are rushing to build infrastructure that concentrates all power in a few



Companies are rushing to build infrastructure that concentrates the power of AI to themselves; automating jobs disproportionately affects the livelihood of lower paid workers.

## 5) Existential Risk:

Climate change driven by industrialization

Nuclear weapons come from advancements in physics

Pandemics spread faster because of globalization

AI will most likely bring about an existential risk.

Look at Daw, 2021 for ethically dubious examples of AI